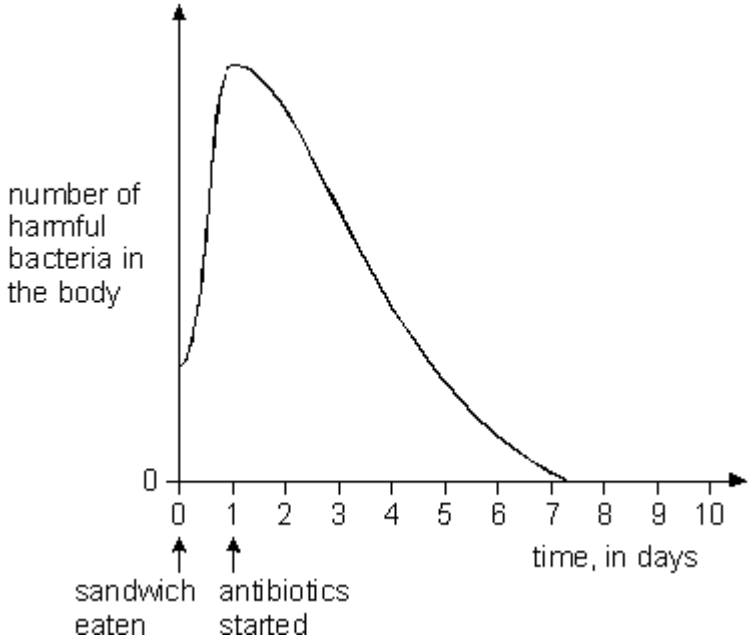
Q1. One evening Jenny and Leah ate chicken sandwiches which had been in their school bags all day. There were harmful bacteria in the food. The next day both girls became very ill. Their doctor gave them antibiotics to take for eight days.

The graph represents how antibiotics affect the number of bacteria in the body.



(a) Use the graph to explain why the girls did **not** become ill until the day after eating the sandwiches.

.....

.....

1 mark

(b) After taking the antibiotics for eight days Jenny was completely better. Explain why she got better.

.....

.....

1 mark

(c) Leah should have taken the antibiotics for eight days. She felt much better after five days and stopped taking the antibiotics. Two days later she felt very ill again.
Use the graph to help you explain why Leah became ill again.

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.....

.....

2 marks

(d) Food will keep longer if it is placed in a refrigerator at 2°C.
Refrigeration does **not** kill bacteria.
What effect does the low temperature have on bacteria?

.....

.....

1 mark
Maximum 5 marks

Mark schemes

Q1.

(a) any **one** from

- there were not enough bacteria in the food **or** body
  accept 'the bacteria had to grow first'
- the bacteria multiplied by the next day

1 (L6)

(b) the antibiotic **or** medicine killed all the bacteria
accept 'the antibiotics got rid of all the bacteria'
***or** 'there were no bacteria left'*

1 (L5)

(c) any **one** from

- antibiotic **or** medicine had not killed all the bacteria
  accept 'not all the bacteria had gone'
- there were still bacteria left alive

1 (L6)

- the bacteria multiplied
  accept 'the population rose again'
  accept 'they could grow again'
  accept 'they reproduced again'

1 (L6)

(d) any **one** from

- it slows down reproduction
  accept 'it stops them reproducing'
  ***or** 'it stops them breeding' **or** 'it stops them multiplying'*
- it is too cold for the bacteria to divide **or** reproduce
  accept 'it stops them growing'
  accept 'slows down growth'
  *do **not** accept 'they are dormant'*
  *do **not** accept 'it freezes them'*

1 (L6)

[5]